

Answers – Emergency Department Worksheet (Parts A–C)

Part A – Structuring the Problem

A1. Decision

How should the Emergency Department be staffed (number of doctors and nurses) so that patients are treated efficiently while maintaining quality of care and controlling costs?

A2. Analytical Questions

How many patients arrive per hour and how does this vary through the day?

How long does treatment take for different types of cases?

How many staff are currently available and what roles do they perform?

How long do patients currently wait before being treated?

What level of waiting time is considered acceptable?

What is the cost of increasing staffing levels?

A3. What analysis would you carry out throw light on the problem?

- Examine arrival patterns over the day (peak times, variability)
- Analyse waiting times at each stage
- Identify bottlenecks (triage vs treatment)
- Compare demand vs staffing capacity
- Test different staffing levels
- Consider prioritisation rules for severe cases
- Use simulation to explore scenarios and trade-offs

A4. What data is required and how will you obtain it?

Data needed

- Patient arrival times and volumes
- Treatment time distributions
- Triage times

- Staffing levels and shift patterns
- Current waiting times and queue lengths
- Mix of case severity

How to obtain

- Hospital records / system logs
- Historical data
- Direct observation
- Staff interviews
- Time-and-motion studies

A5. Factors affecting performance

- Patient arrival patterns (randomness and peak times)
- Variation in treatment duration
- Number of doctors available
- Number of nurses available
- How staff are allocated to tasks
- Whether some patients need faster treatment
- Interruptions to treatment
- Physical space constraints
- Staff fatigue or shift patterns

A6. Possible system sketch description

Patients arrive randomly, wait in a queue, may first be seen by a nurse for initial assessment, then wait to see a doctor for treatment. Some patients may take longer than others. Queues can form at multiple stages. After treatment, patients leave the system.

Part B – Predictions

B1. Effect of increasing staffing

Waiting times should generally decrease as staffing increases, but the improvement will likely level off after a point. If other parts of the process become bottlenecks, adding more staff may not reduce waiting further.

B2. Reasons delays may still occur

- Random surges in patient arrivals
- Very long or complex cases blocking staff
- Bottlenecks at another stage (e.g., assessment or paperwork)
- Uneven distribution of work
- Limited space or equipment

B3. Possible improvements

- Increase number of doctors
- Adjust number of nurses
- Change how staff are allocated
- Introduce a system to handle urgent cases first
- Improve scheduling across the day
- Separate simple and complex cases

Part C – Observations During Modelling

C1. Simple model unrealistic because:

- All patients treated as identical
- No variation in treatment time
- No initial assessment stage
- No prioritisation
- No interruptions
- No variability in arrivals

C2. Additional features for realism

- Different types of patients with different treatment times
- Multiple staff roles (nurses and doctors)
- Random arrival patterns
- Queues at different stages
- Priority for more serious cases

C3. Effect of differing case complexity

If some cases take longer, they can block staff and increase waiting times for others. Queues become more variable and unpredictable, and average waiting time may increase.

C4. Possible prioritisation rule

Patients could be assigned a priority level based on severity. More serious cases would be treated before less serious ones. In extreme situations, treatment of a less urgent case might be paused to treat a more urgent one.

We would probably want to skip triage in critical cases as well.